SPRINT 3: 6th may- 10th june.

- **Sprint Goal:** Build the first single-agent learning prototype, the agent should explore the rescue grid autonomously, use a clear state/action/reward representation, learn from repeated episodes, and produce basic evaluation results across different scenarios.
- **Product Backlog Items:**

Priority	Item	Estimate
1	Implement autonomous exploration baseline: Agents should start exploring without manual movement decisions.	18h
2	Define state-space data structure and reward logic: The learning system needs a clear representation of states, actions, transitions, and rewards.	15h
3	Implement single-agent rescue learning logic and evaluate [1 part]: A single agent should learn or improve behavior before adding multiple agents, the team should evaluate different scenarios.	15h
4	Implement single-agent rescue learning logic and evaluate [2 part]: A single agent should learn or improve behavior before adding multiple agents, the team should evaluate different scenarios.	15h

- **Sprint Backlog and Task assignment:**

Person	Tasks	Hours
xxx	Implement autonomous exploration baseline: Create a non-learning baseline agent that explores without manual movement decisions. It should choose valid moves, avoid walls, keep track of visited cells, and provide a simple comparison point for later learning. Add tests for deterministic behavior and valid movement use.	18h
xxx	Define state-space data structure and reward logic: Design the state representation, action format, transition structure, and reward rules. Include rewards for finding Target A/B, penalties for invalid moves or repeated cells, and episode termination conditions. Add tests and documentation for the learning contract.	15h

xxx	Implement single-agent rescue learning logic and evaluate [1 part]: Q-table and training loop, Implement the first learning loop for one agent using the state/action/reward structure. Store and update Q-values, choose actions with an exploration strategy, run repeated episodes, and save basic training metrics. Add tests for Q-value updates and episode flow.	15h
xxx	Implement single-agent rescue learning logic and evaluate [2 part]: Build evaluation tools for the trained single agent. Run the agent across multiple seeds/scenarios, compare it against the baseline, collect metrics such as success rate, steps taken, targets found, and total reward, and prepare output for visualization or reports.	15h
Everyone	Make sure visualitation and runner.py, api.py is still working.	9h

- **Remaining capacity:**
  4 people x 18h = 72h total.
  Planned work= 63h.
  Buffer = 9h for code review, meetings, and general fixes and communication.